

Subject-Level Calibration Heterogeneity is Stable Across LLM Families and Scales: A Statistical Audit of MMLU

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Motivation

- Calibration is usually reported as one aggregate scalar (ECE), which can hide *where* miscalibration concentrates:
- $$\text{ECE} = \sum_b \frac{|B_b|}{n} |\text{acc}(B_b) - \text{conf}(B_b)|$$
- The analogy:** one accuracy number across demographic groups would not pass as a fairness audit — the same logic applies to calibration.
 - Where** is calibration good and bad, and how much?
 - Is that pattern consistent** across models?
 - What drives it** (question features, model scale)?

Data & Methods

- Data:** MMLU — 14,042 four-option questions across 57 subjects nested in 4 domains (STEM, Social Sciences, Humanities, Other); subject sizes 100–1,534 (median 173).
- Confidence:** renormalized softmax over the four answer tokens:
$$c_i = \max_{a \in \{A, B, C, D\}} \frac{e^{\ell_a}}{\sum_{a'} e^{\ell_{a'}}}$$
- Models:** GPT-4o-mini, Llama-3-8B-Instruct, and Qwen-2.5-{0.5B, 1.5B, 3B, 7B, 14B}-Instruct. Outcome is the signed *calibration gap* $\text{gap}_i = c_i - y_i$, $y_i \in \{0, 1\}$ (positive under overconfidence).
- Model:** three-level mixed model (questions $\rightarrow$ subjects $\rightarrow$ domains):
$$\text{gap}_{ijk} = \mu + \mathbf{x}_{ijk}^T \boldsymbol{\beta} + u_k + v_{jk} + \varepsilon_{ijk},$$
fit via REML.
- Testing:** per-subject one-sample t -tests with Benjamini–Hochberg FDR control at $\alpha = 0.05$; cross-model Spearman correlations of subject-level ECE with bootstrap CIs.

Structure Exists, But Is Modest

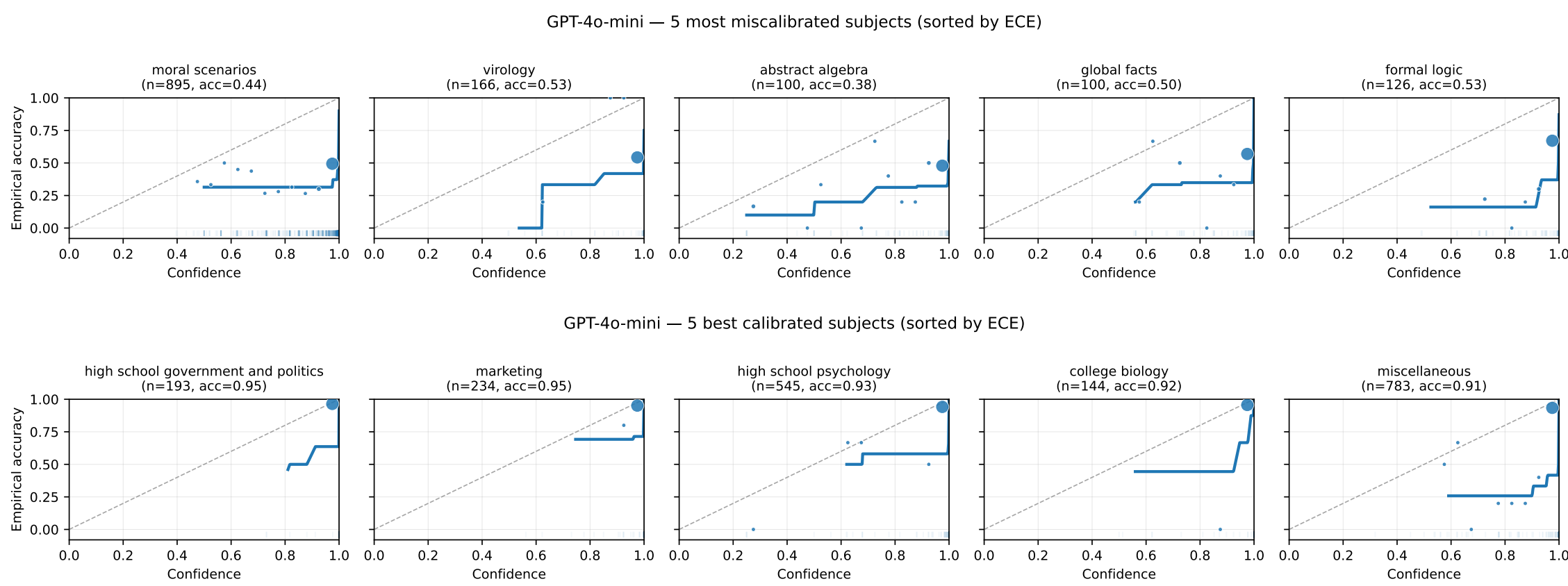
- Most calibration-gap variance is at the question level (92.0–98.5%); domain variance collapses to zero.
- Between-subject ICC spans nearly $5\times$ across models — small in absolute terms. . .
- . . . but (next panel) large enough to organize a $3.8\text{--}13.0\times$ range in subject-level miscalibration.

Model	$\sigma_{\text{subject}}^2$	$\text{ICC}_{\text{subject}}$
GPT-4o-mini	0.013	0.079
Llama-3-8B	0.004	0.020
Qwen-2.5-0.5B	0.004	0.018
Qwen-2.5-1.5B	0.003	0.017
Qwen-2.5-3B	0.011	0.051
Qwen-2.5-7B	0.009	0.049
Qwen-2.5-14B	0.010	0.059

$\text{ICC}_{\text{subject}}$: share of total calibration-gap variance at the subject level. Domain variance is estimated at zero for all models.

Where Calibration Fails

- Every model is overconfident on average:** intercept positive for all seven models, 0.18–0.31 (Guo et al. 2017; Kadavath et al. 2022).
- But the failure is not uniform.** virology and professional law: top-10 worst-calibrated for every model above 1B parameters.
- College-level technical cluster (abstract algebra, college mathematics, machine learning): top-5 for four of six.
- marketing and high school psychology: best-calibrated for every model.
- Specialization level tracks miscalibration: high-school subjects show systematically smaller gaps than their college-level counterparts.



GPT-4o-mini reliability curves: the five most miscalibrated (top) and five best-calibrated (bottom) subjects. Solid: isotonic fit on observed support; markers: 20-bin empirical accuracy sized by count; dashed: perfect calibration.

Magnitude, Not the Binary Count

For each of the 57 subjects s we test whether its mean calibration gap is zero,

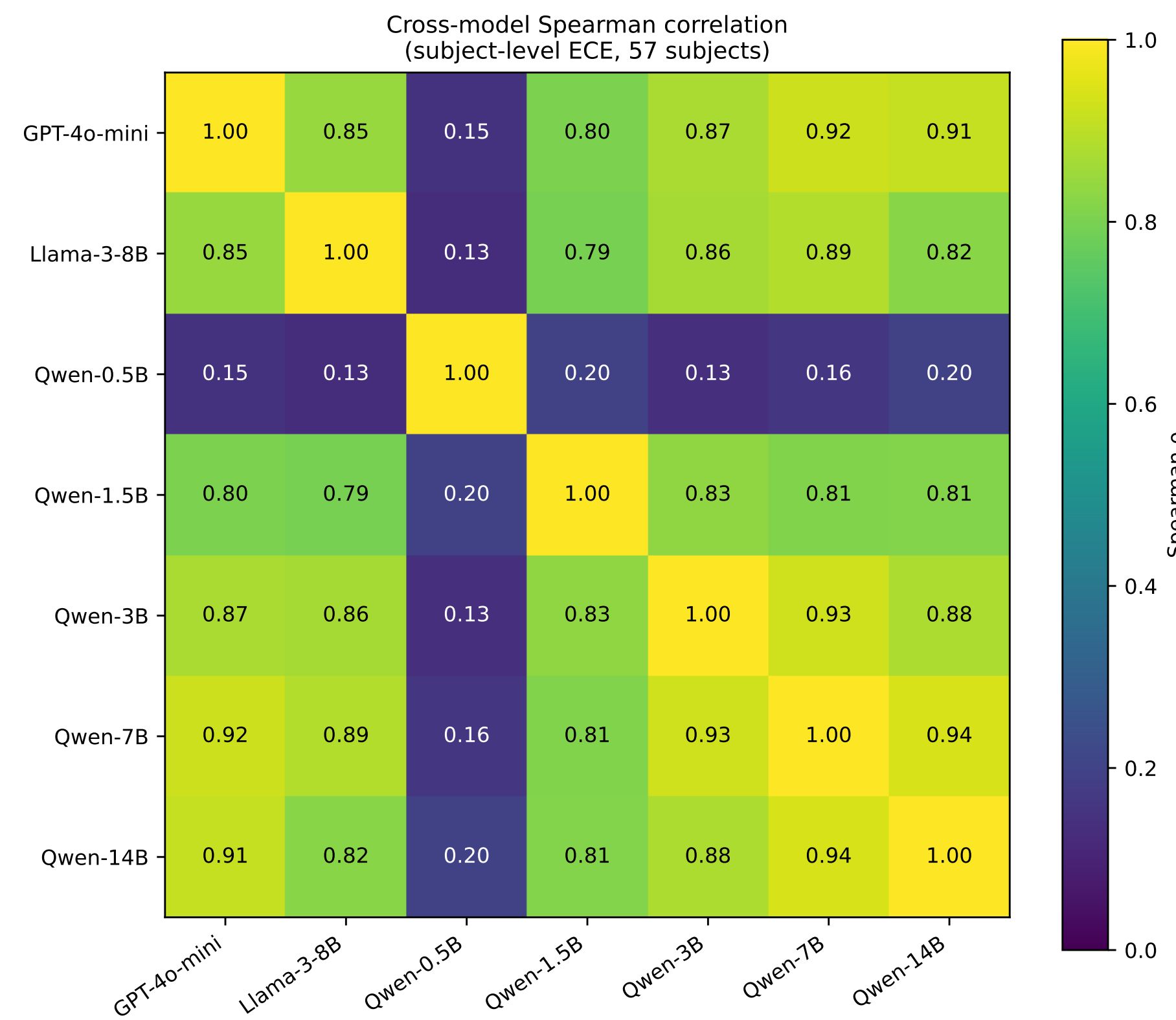
$$H_0 : \mathbb{E}[\text{gap}_i \mid \text{subject } s] = 0 \quad \text{vs.} \quad H_1 : \mathbb{E}[\text{gap}_i \mid \text{subject } s] \neq 0,$$

via a two-sided one-sample t -test, controlling FDR at $\alpha = 0.05$ via Benjamini–Hochberg (1995).

- Rejections are near-universal (the five largest models reject all 57 subjects) — with this much data and a large global overconfidence effect, that’s expected, *not informative*.
- The real finding is the *magnitude*: best- to worst-calibrated gap ranges span $3.8\text{--}13.0\times$ across the six larger models.
- The *same* subjects sit at each extreme, every time.
- Robustness:** ECE rankings are stable across bin counts (Spearman ≥ 0.97 vs. the 20-bin baseline) and agree with a strictly proper scoring rule, the Negative Log Calibration Score (Spearman ≥ 0.79); 2- and 3-level ICC estimates match.

Stable Across Models and Families

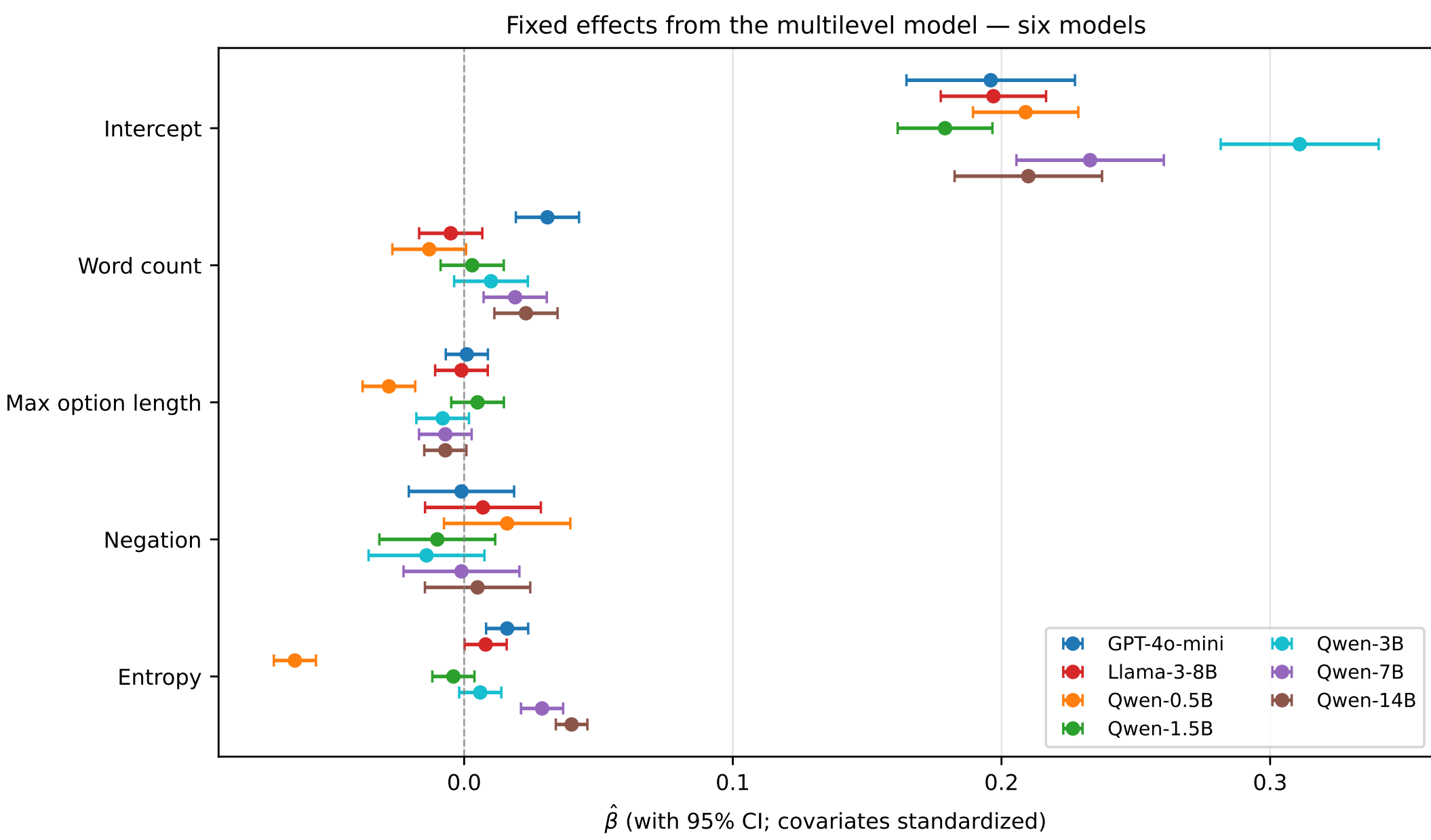
- All 15 pairwise Spearman correlations among the six larger models fall in $[0.80, 0.94]$.
- Cross-family GPT-4o-mini vs. Qwen-7B/14B (0.92/0.91) is as strong as the best within-family pair (Qwen-7B vs. 14B, 0.94).
- The same subjects are hard *regardless of architecture or scale*.
- Qwen-0.5B is a conspicuous exception ($\rho \in [0.13, 0.20]$), driven by $10.6\times$ positional bias toward D — a floor below which our elicitation protocol breaks down.



Spearman rank correlation of subject-level ECE across the seven models (57 common subjects).

What Predicts the Gap

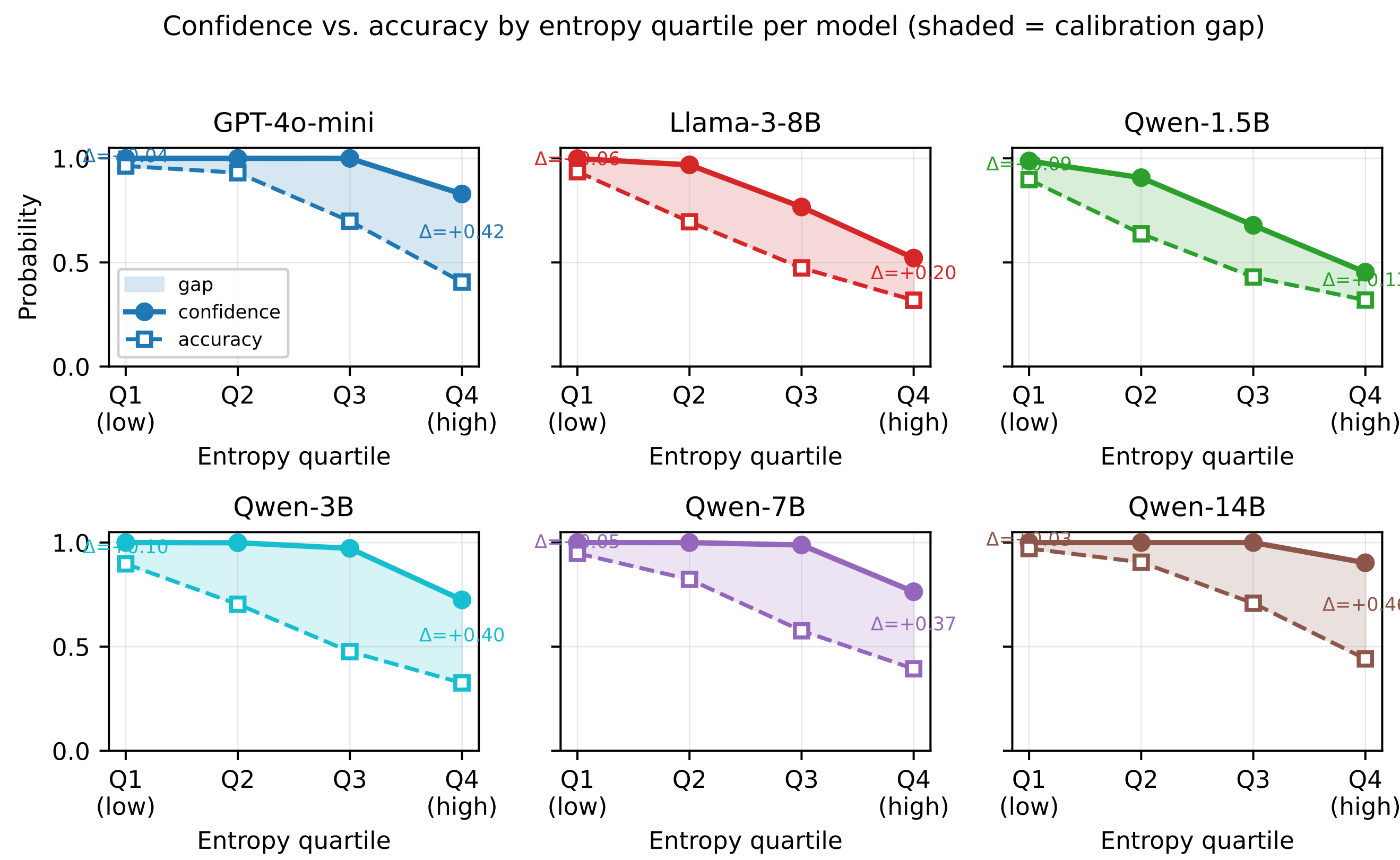
- Word count positively predicts overconfidence for GPT-4o-mini and the larger Qwens.
- Negation and option length: mostly inert.
- Most informative pattern: the **entropy** coefficient (next panel).



Fixed-effect estimates ($\pm 95\%$ CI) on the standardized calibration-gap scale. The entropy coefficient shows a clear sign progression across the Qwen scaling sweep.

The Gap Grows With Scale

- Entropy H_i of the A/B/C/D distribution proxies item difficulty. Naively, higher entropy should *shrink* the gap (top-1 confidence is mechanically lower).
- Instead, within the matched-precision Qwen sweep (1.5B fp16, 3B, 7B, 14B), the entropy coefficient grows monotonically:
$$-0.004 \rightarrow +0.006 \rightarrow +0.029 \rightarrow +0.040.$$
- At 7B/14B, confidence stays near saturation across the first three entropy quartiles while accuracy keeps dropping, so the gap widens on the model’s hardest items (Q1 vs. Q4 gap: $+0.05$ vs. $+0.37$ at 7B; $+0.03$ vs. $+0.46$ at 14B).
- Between-subject ICC also jumps from 1.5B to 3B ($0.017 \rightarrow 0.051$) and is flat thereafter — consistent with RLHF-induced confidence saturation (Kadavath et al. 2022; Nakkiran et al. 2025).



Confidence (solid) and accuracy (dashed) by entropy quartile; shaded area is the calibration gap.

Why Subject-Level, Not Model-Level

- A single global fix (e.g. temperature scaling) cannot work here: one scalar T tuned for the worst subjects (professional law, virology, gap up to 0.487) would badly overcorrect well-calibrated subjects like marketing (gap 0.037) — the $13\times$ spread is too wide for any single correction.
- Because the same subjects are hard *across every model family and scale we tested*, this is a property of the *task*, not a training-pipeline quirk — so the fix has to live at the subject level too.
- Practical upshot:** the recurring worst-calibrated subjects double as a portable pre-deployment checklist — flag likely miscalibration on a *new* model without re-running the full 57-subject audit.

Takeaways

- The unit of analysis is the subject, not the model.** Aggregate ECE hides a $3.8\text{--}13.0\times$ range in miscalibration magnitude that a single scalar cannot see.
- Miscalibration is task-driven.** Subject rankings are stable across three families and six scales ($\rho \in [0.80, 0.94]$) — model-level retraining is unlikely to resolve it.
- Open question:** *why* these subjects are hardest (abstractness, training-data rarity, or something else) remains unresolved.

References

Benjamini & Hochberg (1995). Controlling the false discovery rate. *JRSS-B* 57(1). Guo et al. (2017). On calibration of modern neural networks. *ICML*. Hendrycks et al. (2021). Measuring massive multitask language understanding. *ICLR*. Kadavath et al. (2022). Language models (mostly) know what they know. arXiv:2207.05221. Nakkiran et al. (2025). Trained on tokens, calibrated on concepts. arXiv:2511.04869. Xiong et al. (2024). Can LLMs express their uncertainty? *ICLR*.